AI-ASSISTED SELF-COACHING SYSTEM FOR KABADDI RAIDERS USING POSE ESTIMATION AND SPLIT-SCREEN VIDEO COMPARISON

A PROJECT REPORT

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ABSTRACT

Kabaddi is a physically demanding sport that requires correct body posture, proper timing, and accurate execution of raiding movements. Traditional Kabaddi training depends largely on experienced coaches, which makes effective self-practice difficult when guidance is not available. In addition, existing training methods do not provide objective analysis or personalized feedback to help players identify their mistakes.

This thesis presents an AI-assisted self-coaching system for Kabaddi raiders using pose estimation and temporal analysis. The system uses a split-screen comparison approach to visually compare a student's performance with a professional reference video. A custom dataset was manually collected from college-level Kabaddi players under real playing conditions, focusing on Toe Touch, Running Hand Touch, and Scorpion Kick movements. Video frames are processed to extract human pose keypoints, which are normalized and converted into joint angle representations. A BiLSTM-based temporal model with attention is used to identify important execution phases. Based on movement similarity and joint deviations, the system provides simple, rule-based feedback to help players improve their technique. This work demonstrates the usefulness of AI-based self-coaching for Kabaddi training.

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LIST OF SYMBOLS AND ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
ASR	Attack Success Rate
BERT	Bidirectional Encoder Representations from Transformers
BLEU	Bilingual Evaluation Understudy
CER	Character Error Rate
CoT	Chain-of-Thought
EMA	Exact Match Accuracy
F1	F1-Score (Harmonic Mean of Precision and Recall)
FNR	False Negative Rate
FPR	False Positive Rate
GUI	Graphical User Interface
HTML	HyperText Markup Language
JSON	JavaScript Object Notation
LDP	Local Differential Privacy
LLM	Large Language Model
MCTS	Monte Carlo Tree Search
NLP	Natural Language Processing
OCR	Optical Character Recognition
RAG	Retrieval-Augmented Generation
T5	Text-to-Text Transfer Transformer
VLM	Vision-Language Model

CHAPTER 1

INTRODUCTION

Kabaddi is a popular indigenous sport that combines strength, agility, speed, and strategic thinking. Raider performance in Kabaddi depends heavily on precise body posture, correct timing, and accurate execution of movements such as Toe Touch, Running Hand Touch, and Scorpion Kick. Small errors in joint coordination or execution phase can significantly affect performance and scoring. Traditionally, Kabaddi training relies on continuous observation and guidance from experienced coaches to correct these errors and improve technique.

However, effective coaching support is not always available, especially during individual practice sessions. Many players practice without expert supervision and are unable to objectively analyze their movements or identify technical mistakes. Existing training methods largely depend on visual observation and subjective feedback, which may vary from coach to coach. Furthermore, there is a lack of automated systems that provide quantitative analysis and personalized feedback specifically for Kabaddi raiders.

Recent advancements in computer vision and deep learning have enabled detailed analysis of human motion using video data. Pose estimation techniques can extract skeletal joint information from videos, allowing movements to be represented in a structured and interpretable form. When combined with temporal modeling techniques, these representations make it possible to analyze motion patterns, identify key execution phases, and compare performance across different players. Such approaches have shown promising results in sports like badminton, gymnastics, and athletics, motivating their

application to Kabaddi.

This thesis proposes an AI-assisted self-coaching system for Kabaddi raiders that leverages pose estimation and temporal modeling for movement analysis. The system uses a split-screen framework to compare a student's execution with professional reference movements. By analyzing pose sequences and identifying important execution phases using deep learning models, the system objectively evaluates performance and provides understandable, rule-based feedback. The proposed approach aims to support effective self-practice, reduce dependence on continuous coaching, and contribute toward intelligent training systems for Kabaddi.

1.1 PROBLEM STATEMENT

Kabaddi raiding performance depends on accurate body posture, proper timing, and correct execution of movements. Traditional Kabaddi training relies heavily on continuous supervision by experienced coaches, making effective self-practice difficult when expert guidance is unavailable. Players practicing independently often struggle to identify posture errors and execution mistakes due to the lack of objective analysis tools. Current training methods are largely subjective and do not provide quantitative evaluation or personalized feedback. Moreover, there is no automated system specifically designed for Kabaddi that can analyze movement execution, identify key temporal phases, and compare student performance with professional standards.

Although pose estimation and deep learning techniques have been successfully applied in other sports, their use in Kabaddi training remains limited. Existing approaches do not effectively capture the temporal dynamics of raiding actions or convert technical errors into understandable corrective feedback.

Therefore, there is a need for an AI-based self-coaching system that can objectively analyze Kabaddi raider movements, compare executions with professional references, and provide clear, interpretable feedback to support effective self-training and skill improvement.

1.2 OBJECTIVE

The primary objective of this study is to develop an intelligent system that supports effective self-practice for Kabaddi raiders. By leveraging pose estimation and deep learning techniques, the work aims to objectively analyze movement execution and identify key performance errors.

- To develop an AI-assisted self-coaching system for analyzing Kabaddi raider movements.
- To collect and construct a custom Kabaddi dataset focusing on key raiding actions.
- To extract human pose keypoints from video data using pose estimation techniques.
- To perform temporal phase detection and identify important execution frames using deep learning.
- To detect posture and joint-level deviations during movement execution.
- To generate human-interpretable, rule-based feedback for technique improvement.
- To support effective self-practice and reduce dependency on continuous coaching guidance.

CHAPTER 2

LITERATURE SURVEY

The literature survey presents a comprehensive review of existing research related to Kabaddi skill analysis, human pose estimation, temporal modeling, and AI-based coaching systems. It highlights prior work on Kabaddi rules and biomechanics, video preprocessing, pose estimation techniques, and deep learning approaches for action analysis. This review helps identify research gaps and motivates the proposed pose-based self-coaching framework for Kabaddi raiders.

2.1 Kabaddi Rules and Skill Analysis

Kabaddi is governed by standardized rules that define raiding techniques, scoring methods, and movement constraints. The official rules published by the International Kabaddi Federation [3] provide a structured understanding of legal raiding actions such as toe touch, hand touch, and kicking techniques. Biomechanical analysis of Kabaddi movements by Dey et al. [1] highlighted the importance of posture control, joint coordination, and execution timing in successful raids. Further motion-based performance analysis by Kumar and Shanmugam [2] emphasized that minor deviations in execution phases can significantly affect raiding efficiency. These studies underline the need for objective, movement-based analysis systems in Kabaddi training.

2.2 Frame Extraction and Video Preprocessing

Frame extraction is a fundamental step in video-based action analysis, enabling frame-level feature extraction and temporal modeling.

Bradski [7] introduced OpenCV as a comprehensive library for video processing tasks such as frame extraction and image manipulation. Practical implementations of video preprocessing and efficient frame handling using OpenCV were further discussed by Rosebrock [8]. Gonzalez and Woods [9] presented foundational image processing techniques that support preprocessing steps such as normalization and enhancement, improving data quality prior to pose estimation.

2.2.1 Pose Estimation Techniques

Pose estimation enables the extraction of skeletal joint information for representing human movements. Shotton et al. [12] demonstrated early work on real-time human pose recognition using depth images. Newell et al. [13] proposed stacked hourglass networks that improved pose estimation accuracy through multi-scale spatial feature learning. Cao et al. [10] introduced real-time multi-person 2D pose estimation using part affinity fields. Lugaresi et al. [11] later proposed MediaPipe Pose, a lightweight and efficient framework suitable for real-time and mobile-based applications.

2.3 Annotation Tools and Techniques

Accurate annotation is essential for supervised learning tasks such as temporal phase detection. The VGG Image Annotator (VIA) tool introduced by Dutta and Zisserman [4] supports manual annotation of images and videos at frame level. LabelMe [5] provides a widely used web-based annotation platform for visual datasets. To improve annotation efficiency, Papadopoulos et al. [6] proposed extreme clicking methods that reduce labeling effort while maintaining annotation accuracy.

2.4 Spatio-Temporal Graph-Based Modeling

Graph-based approaches effectively model skeletal data by capturing spatial joint connections and temporal dynamics. Yan et al. [14] proposed Spatial Temporal Graph Convolutional Networks (ST-GCN) for skeleton-based action recognition. Shi et al. [15] introduced adaptive graph convolution techniques to dynamically learn joint relationships. Li et al. [16] further enhanced graph modeling by incorporating actional-structural representations.

2.5 Temporal Phase Detection and Sequence Learning

Temporal modeling is essential for understanding sequential motion patterns. Long Short-Term Memory (LSTM) networks introduced by Hochreiter and Schmidhuber [17] address long-term dependency issues in sequential data. Bidirectional LSTM models proposed by Graves and Schmidhuber [18] improve temporal learning by incorporating both past and future context. Bahdanau et al. [19] introduced attention mechanisms that assign importance to critical frames, enabling effective temporal phase detection.

2.6 Similarity Learning and Skill Assessment

Similarity learning techniques are widely used to compare movement executions. Siamese networks introduced by Koch et al. [20] enable learning similarity metrics using paired inputs. Discriminative similarity learning for verification tasks was explored by Chopra et al. [21]. Triplet-based metric learning was proposed by Hoffer and Ailon [22]. Action quality assessment approaches [23, 24, 25] quantify execution quality by analyzing motion consistency and deviations. Explainable and neural-symbolic learning methods [26, 27] support interpretable feedback generation for coaching systems.

2.7 SUMMARY

The literature survey examined studies related to Kabaddi biomechanics, pose estimation, video preprocessing, temporal modeling, and AI-based coaching systems. Existing research highlights the importance of posture, joint coordination, and execution timing in Kabaddi raiding performance, while also revealing the absence of objective and automated training support in traditional coaching methods. Prior work in pose estimation and deep learning demonstrated the effectiveness of skeletal representations for analyzing human movements. Temporal modeling techniques such as BiLSTM and attention mechanisms were shown to successfully identify key execution phases. Similarity learning and action quality assessment methods enabled quantitative comparison of movement executions. However, the application of these techniques to Kabaddi training is limited. This research addresses the identified gap by proposing a pose-based self-coaching system for Kabaddi raiders.

CHAPTER 3

SYSTEM DESIGN AND IMPLEMENTATION

3.1 System Architecture

The proposed system is an AI-assisted pose-based skill assessment framework designed to support self-coaching for Kabaddi raiders. As shown in the system architecture diagram, the framework integrates video processing, pose estimation, temporal modeling, similarity learning, and feedback generation into a unified pipeline. The system analyzes student raider performances by comparing them with professional reference executions and provides objective, human-understandable coaching feedback.

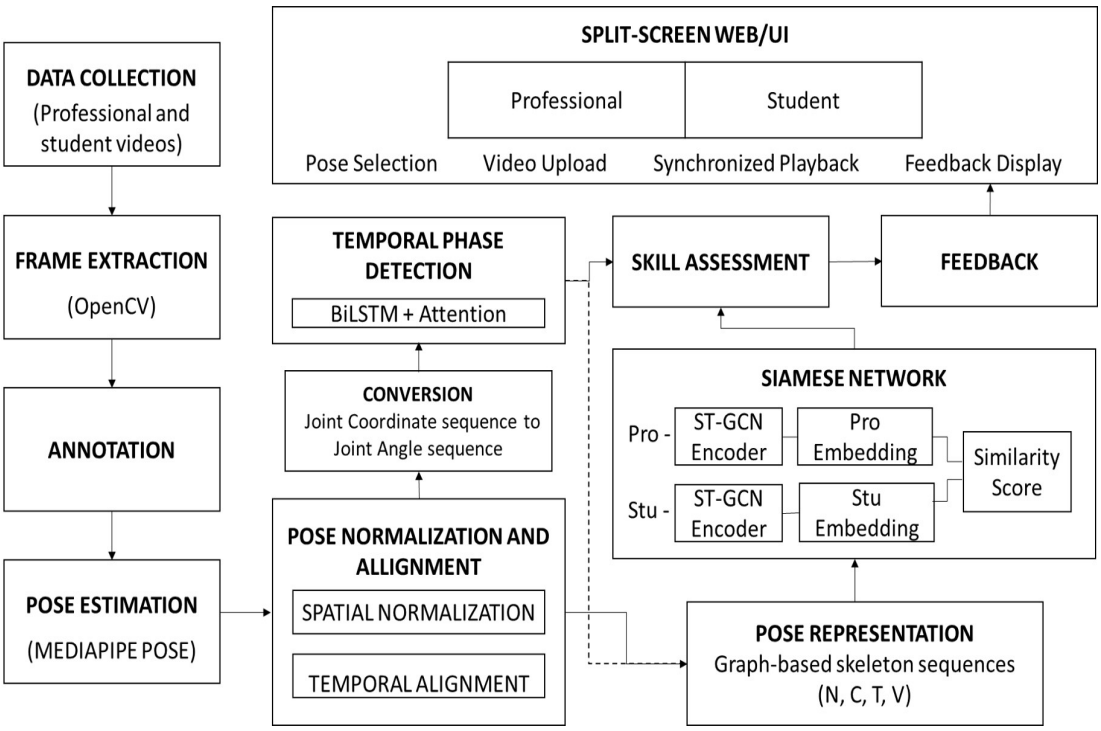

Figure 3.1: Pose-based Skill Assessment Network

The architecture consists of multiple interconnected modules, starting from data collection and preprocessing, followed by pose extraction

and normalization. Temporal phase detection is employed to identify critical execution frames, while a Siamese network based on ST-GCN encoders evaluates similarity between student and professional movements. The final stages perform skill assessment and feedback generation, which are presented through a split-screen user interface for effective self-practice.

The major modules implemented in this architecture are listed below:

- Data Collection Module
- Annotation Module
- Temporal Phase Detection Module
- Siamese Network Module
- Skill Assessment Module
- Feedback Generation Module

3.2 Data Collection Module

The data collection process Fig 3.2 forms the foundation of the proposed system. A custom Kabaddi dataset was manually collected under real playing conditions at the college Kabaddi ground. The dataset consists exclusively of raider-only videos to eliminate interference from defenders and background motion.

Two categories of videos are collected: professional reference executions and student performance videos. All videos are categorized action-wise into three fundamental raiding movements, namely Toe Touch, Running Hand Touch, and Scorpion Kick. Each video captures a complete execution of a single raid

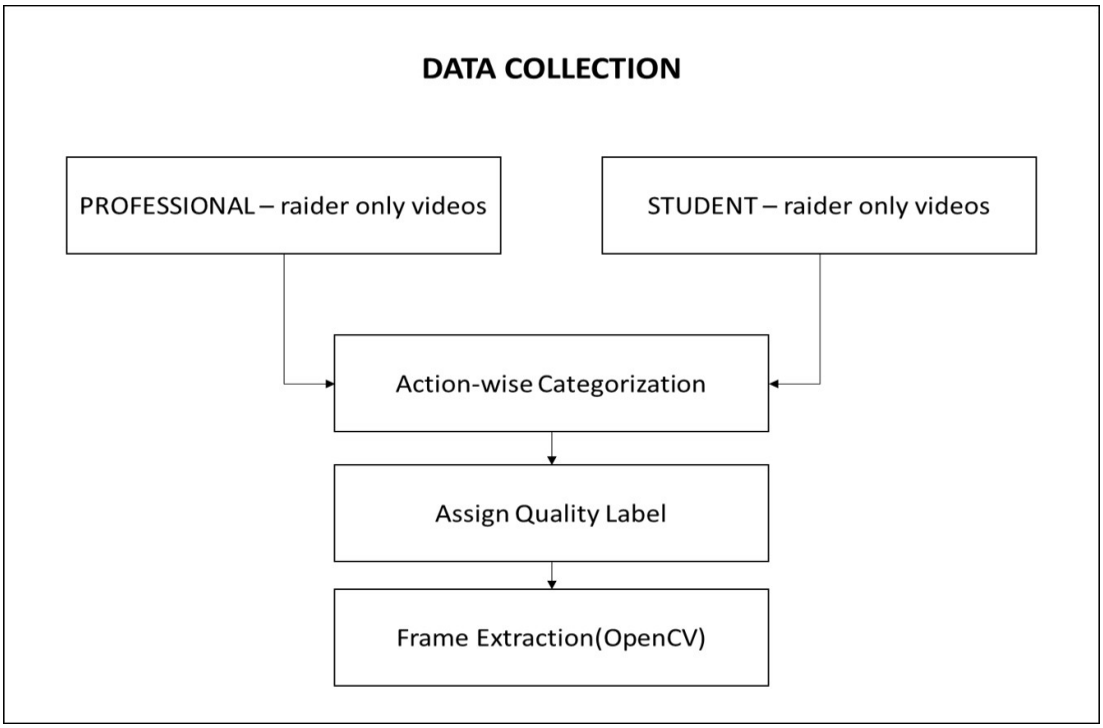

Figure 3.2: Module 1: Data Collection

action. To ensure robustness, variations in execution speed, posture, and player style are included. The collected videos are processed using OpenCV to extract frame sequences for further analysis.

3.3 Annotation Module

The annotation module Fig 3.3 is responsible for generating ground-truth labels required for supervised learning. Annotation is performed manually using the VIA (VGG Image Annotator) tool on the extracted frame sequences. Frames are viewed sequentially, and each video is labeled with the corresponding raiding movement type.

For each action, the execution phase is identified by marking the start and end frames of the key movement segment. Frame numbers are recorded, and a quality label is assigned based on execution correctness. The annotated data

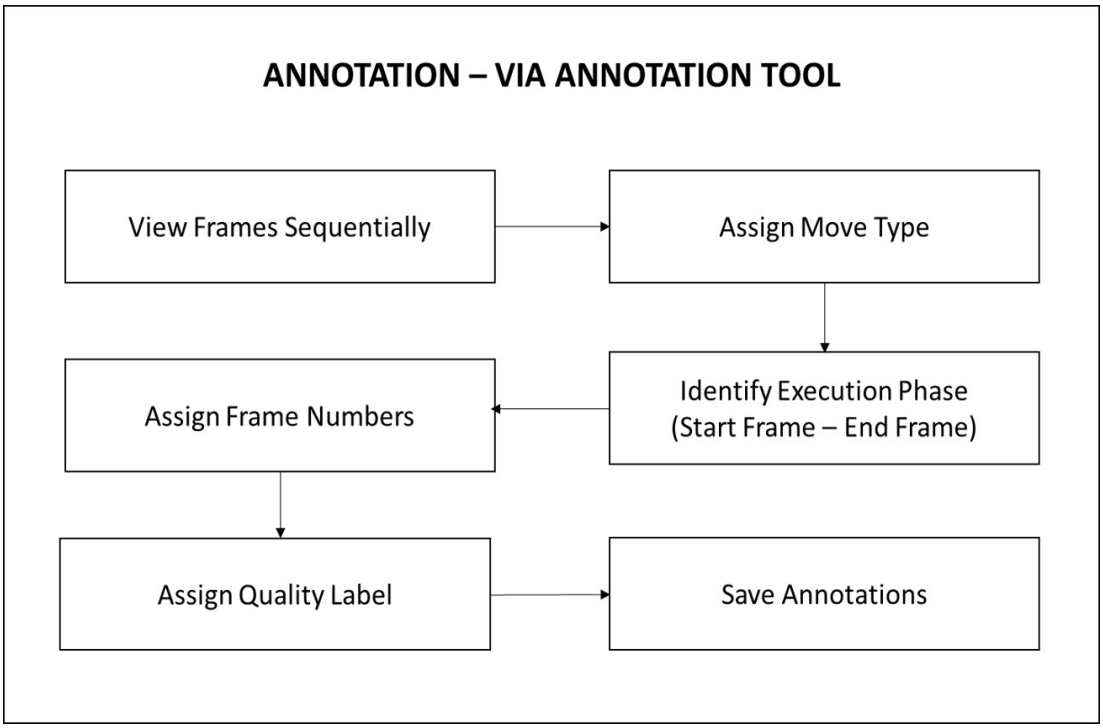

Figure 3.3: Module 2: Annotation Process

is stored in a structured format and used during training and evaluation of the temporal phase detection and similarity learning modules.

3.4 Temporal Phase Detection Module

The temporal phase detection module Fig 3.4 identifies the most important execution frames within a raiding action. The input to this module is a joint angle sequence obtained after pose normalization and coordinate-to-angle conversion. Since video sequences vary in length, sequence padding is applied to maintain uniform input dimensions.

A Bidirectional Long Short-Term Memory (BiLSTM) network is employed to capture temporal dependencies in both forward and backward directions. Outputs from the forward and backward LSTM layers are concatenated to form a comprehensive temporal feature representation. An attention layer assigns

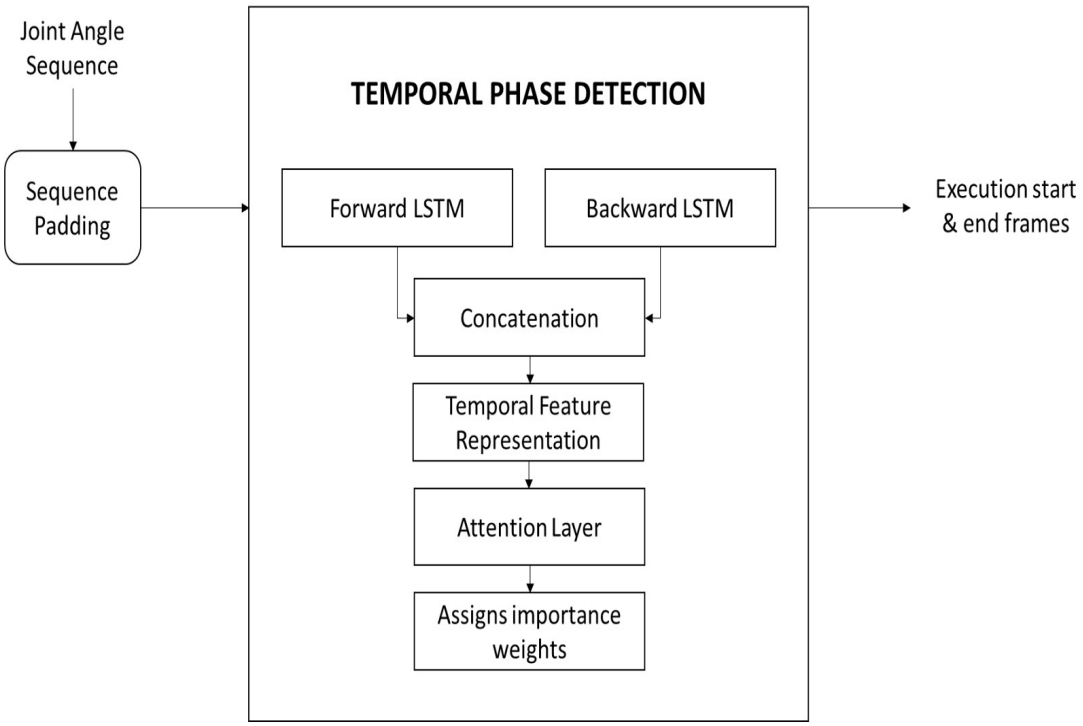

Figure 3.4: Module 3: Temporal Phase Detection using BiLSTM and Attention

importance weights to individual frames, enabling the model to focus on critical execution phases. The module outputs the execution start and end frames along with frame-level importance scores.

3.5 Siamese Network Module

The Siamese Network module Fig 3.5 is designed to compare student performances with professional reference executions. The input to this module is a graph-based pose representation structured as a spatio-temporal tensor (N, C, T, V), where joints are represented as nodes and skeletal connections as edges.

Professional and student pose sequences are processed through identical ST-GCN encoders consisting of spatial convolution, temporal convolution, and stacked ST-GCN blocks. Global pooling is applied to

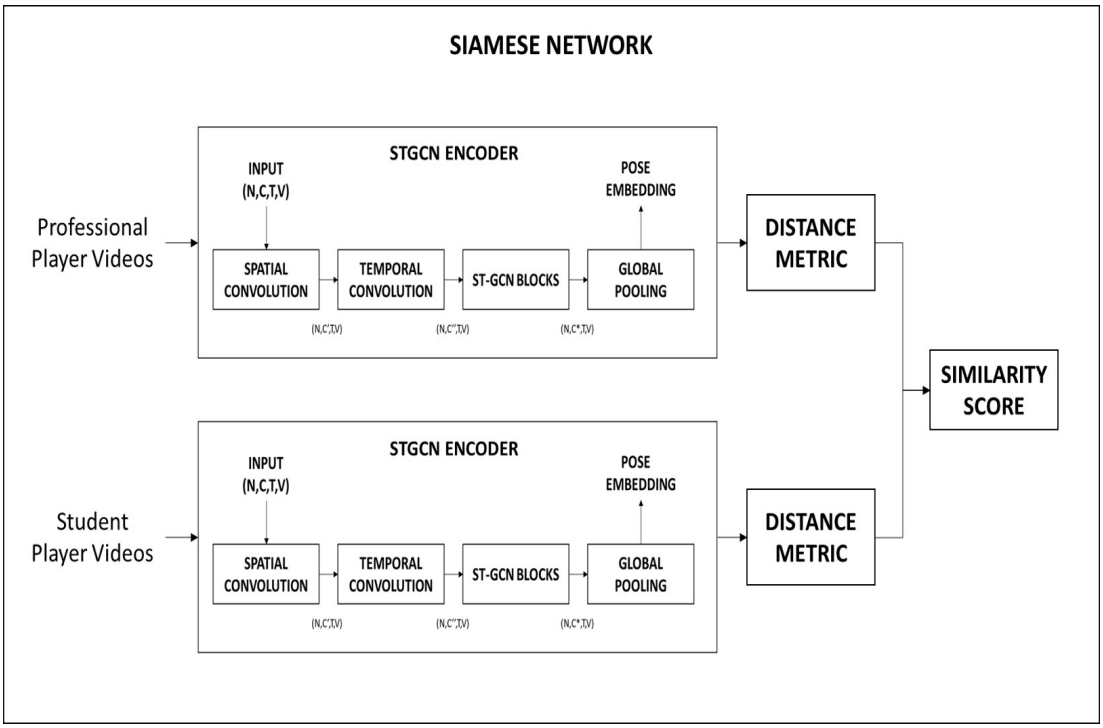

Figure 3.5: Module 4: Siamese Network based Skill Comparison

obtain compact pose embeddings. A distance metric computes the similarity between professional and student embeddings, producing a similarity score that quantifies execution quality.

3.6 Skill Assessment Module

The skill assessment module Fig 3.6 integrates outputs from the Siamese network and temporal phase detection module. Inputs include similarity scores, attention weights, and pose embeddings of professional and student executions. Based on these inputs, the system identifies joint-level deviations and error-prone temporal segments.

Performance is categorized into predefined skill levels such as Good, Average, or Needs Improvement. The module also highlights joints with high deviation and identifies frame ranges where execution errors occur, enabling

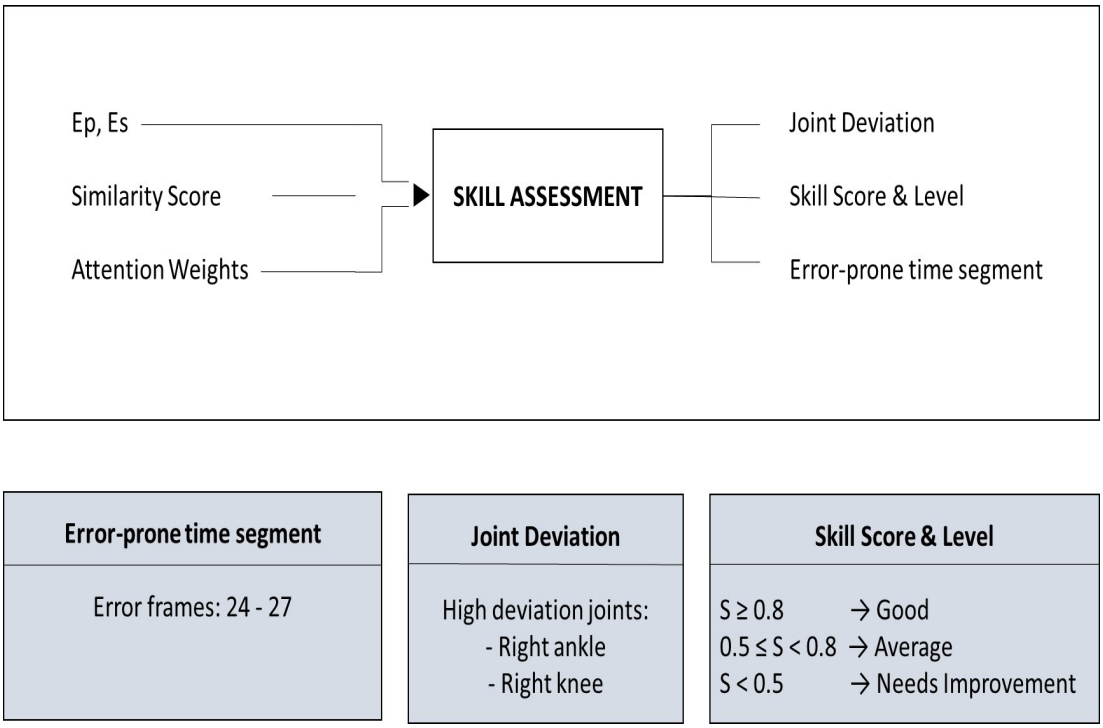

Figure 3.6: Module 5: Skill Assessment and Error Analysis

structured interpretation of the assessment results.

3.7 Feedback Generation Module

The feedback generation module Fig 3.7 converts technical assessment outputs into human-understandable coaching feedback. A rule-based knowledge base maps joint deviations and angle thresholds to corrective suggestions. Confidence-aware feedback is generated based on the assessed skill level to ensure constructive and motivational guidance.

Feedback aggregation and formatting combine multiple observations into a concise message. The final feedback provides actionable suggestions related to posture correction, joint alignment, and execution timing. This feedback is displayed through a split-screen interface alongside synchronized professional and student videos.

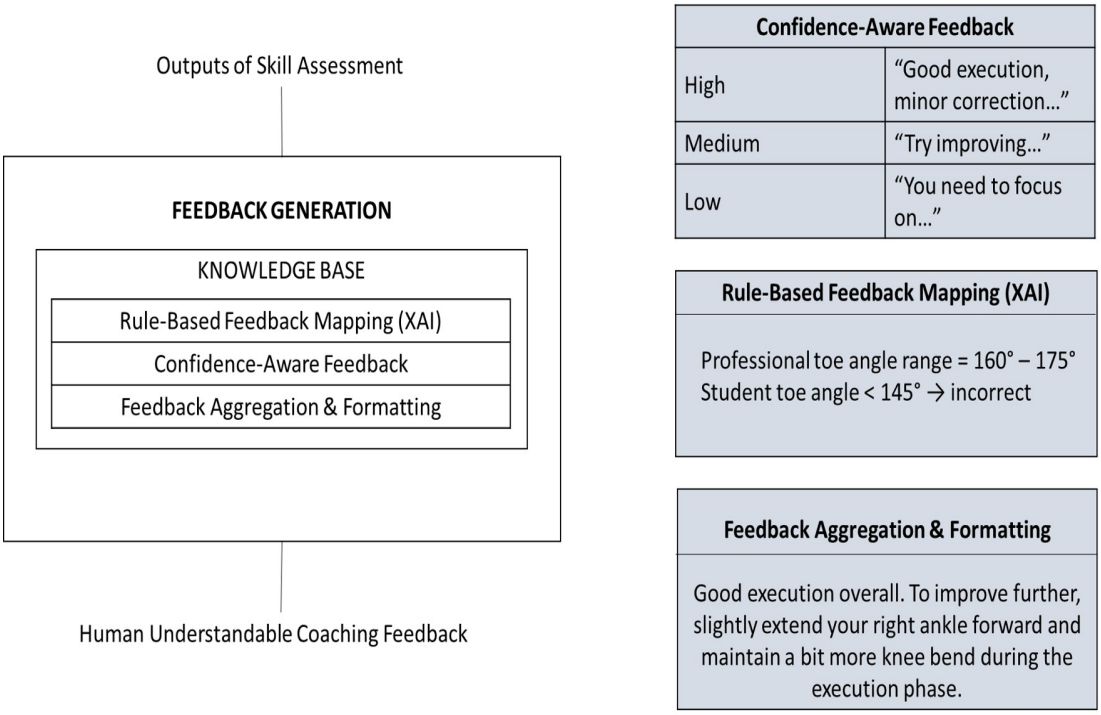

Figure 3.7: Module 6: Rule-Based Feedback Generation

3.8 Summary

This chapter presented the system design and implementation of the proposed pose-based skill assessment framework for Kabaddi raiders. The system integrates video preprocessing, pose estimation, temporal modeling, similarity learning, and explainable feedback generation into a unified architecture. By combining deep learning techniques with rule-based interpretation, the proposed framework enables objective skill evaluation and effective self-coaching for Kabaddi training.

CHAPTER 4

IMPLEMENTATION DETAILS

This chapter describes the implementation of the proposed pose-based analysis system for Kabaddi raiders. Implementation focuses on data preparation, pose extraction, temporal modeling, and key execution phase identification. Advanced modules such as similarity learning and feedback generation are planned for future work.

4.1 Dataset Collection

A custom Kabaddi dataset was manually collected from college-level players under real playing conditions. Only raider actions were recorded to avoid background interference. The dataset includes three raiding movements: Toe Touch, Running Hand Touch, and Scorpion Kick. Each video contains a single complete execution with variations in posture and speed.

4.2 Frame Extraction

Video data is converted into frame sequences using the OpenCV library. Frames are extracted at a consistent frame rate and stored in an organized directory structure based on movement type. This enables frame-level analysis required for pose estimation and temporal modeling.

4.3 Pose Estimation

Pose estimation is performed using MediaPipe Pose to extract 2D skeletal joint keypoints for each frame. The extracted joint coordinates represent

the posture and movement of the raider over time and serve as input for subsequent processing stages.

4.4 Pose Normalization and Joint Angle Conversion

Spatial normalization is applied to reduce variations in body size and position across different players. The normalized joint coordinates are then converted into joint angle representations, which effectively capture relative joint motion and reduce sensitivity to camera viewpoint differences.

4.5 Annotation

Manual annotation is carried out using the VIA (VGG Image Annotator) tool to label execution phases within each video. These annotations provide ground truth for supervised temporal learning and enable accurate identification of key execution frames.

4.6 Temporal Phase Detection

Temporal phase detection is implemented using a Bidirectional Long Short-Term Memory (BiLSTM) network with an attention mechanism. The model processes joint angle sequences to learn temporal dependencies and assigns importance scores to individual frames. Frames with higher attention weights are identified as key execution phases.

4.7 Summary

The implemented pipeline successfully transforms raw Kabaddi videos into temporally refined pose representations and identifies key execution

phases. This work establishes a strong foundation for skill comparison and feedback generation in future phases of the project.

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